

ACIDS, BASES AND SALTS

Student Learning Pack

1. Metadata

Topic	Acids, Bases and Salts
Level	Class X
Lesson Length	60 minutes

Scope Note

Focus on classification of salts, pH behaviour based on parent acid/base strength, and interpretation of experimental observations.

2. Learning Objectives

- LO1: Identify the acids and bases used to form given salts and classify salts into families.
- LO2: Explain how the strength of the parent acid and base determines the pH of the salt solution.
- LO3: Interpret experimental pH data to categorize salts as acidic, basic or neutral.
- LO4: Describe the pH scale and its significance for classifying solutions.
- LO5: Recognise sodium chloride as a neutral salt and relate it to everyday uses.

3. Prerequisite Knowledge

- Understanding that acids release H^+ ions and bases release OH^- ions in aqueous solution.
- Difference between strong and weak acids/bases.
- Use of litmus paper and pH paper to test acidity or basicity.
- Basic idea that salts are formed by neutralisation of an acid with a base.

4. Key Terms

Term	Definition
Salt	A compound formed when an acid reacts with a base, consisting of the cation from the base and the anion from the acid.
Acid	Substance that releases H^+ ions in water.
Base	Substance that releases OH^- ions in water.
Strong acid	An acid that dissociates completely in water, e.g., HCl.
Weak base	A base that does not dissociate completely in water, e.g., ammonia.
pH	A numerical scale (0-14) that indicates the concentration of hydrogen ions; lower values are acidic, higher values are basic.
Neutral salt	A salt formed from a strong acid and a strong base; its aqueous solution has $pH \approx 7$.

Term	Definition
Acidic salt	A salt whose solution has $\text{pH} < 7$, typically formed from a strong acid and a weak base.
Basic salt	A salt whose solution has $\text{pH} > 7$, typically formed from a weak acid and a strong base.
Salt family	Group of salts sharing the same positive ion (cation) or the same negative ion (anion).

5. Key Points

- Salts sharing the same cation or anion belong to the same family.
- The pH scale (0-14) indicates whether a solution is acidic, neutral, or basic.
- A salt's pH depends on whether its parent acid and base are strong or weak.
- Students test solubility, observe litmus colour change, and record pH of various salts.
- Sodium chloride, formed from a strong acid and strong base, is neutral and widely used.

6. Real-World Applications

Sodium Chloride in Food Preservation

Common salt inhibits microbial growth by reducing water activity, keeping food safe and tasty.

Think about: Why does adding NaCl to meat help it stay fresh longer?

Suggested time: 3 minutes

pH of Household Cleaning Solutions

Many cleaning agents are either acidic (e.g., vinegar) or basic (e.g., bleach) to enhance their effectiveness.

Think about: A bathroom cleaner turns blue litmus red. What does this indicate about its pH and why is that useful for cleaning?

Suggested time: 4 minutes

7. Core Notes

Salt Families and Identification

A salt family is defined by a common positive (cation) or negative (anion) radical. For example, NaCl and Na_2SO_4 are both sodium salts because they contain the Na^+ ion. NaCl and KCl belong to the chloride family because they contain the Cl^- ion. In Activity 2.14 the listed salts - sodium chloride, potassium nitrate, aluminium chloride, zinc sulphate, copper sulphate, sodium acetate, sodium carbonate and sodium hydrogencarbonate - can be grouped by their cations (Na^+ , K^+ , Al^{3+} , Zn^{2+} , Cu^{2+}) or anions (Cl^- , NO_3^- , SO_4^{2-} , CH_3COO^- , CO_3^{2-} , HCO_3^-).

The pH Scale and Solution Classification

pH measures the concentration of hydrogen ions (H^+) in a solution. A pH of 7 is neutral; values below 7 are acidic, and values above 7 are basic. Living organisms function best within a specific pH range. Litmus paper provides a quick test: red litmus turning blue shows a basic solution ($\text{pH} > 7$).

How Parent Acid-Base Strength Determines Salt pH

- Strong acid + strong base → neutral salt ($\text{pH} \approx 7$).
- Strong acid + weak base → acidic salt ($\text{pH} < 7$).
- Strong base + weak acid → basic salt ($\text{pH} > 7$).

These rules follow from the relative tendency of the conjugate acid or base to donate or accept H^+ ions. For example, sodium chloride (from $\text{HCl} + \text{NaOH}$) is neutral, while salts formed from a weak acid such as acetic acid and a strong base like NaOH are basic.

Activity 2.14: Solubility, Litmus Test and pH Measurement of Salts

Steps:

1. Collect the eight listed salts.
2. Dissolve each in distilled water and note whether it dissolves (solubility).
3. Dip red and blue litmus strips into each solution; record colour change.
4. Use pH paper to determine the pH value.
5. Complete Table 2.4 with the observed pH, the acid and base that formed each salt, and classify the salt as acidic, basic or neutral.

The activity reinforces the relationship between parent acid/base strength and the observed pH.

Common Salt (NaCl) - A Neutral Salt and Its Everyday Significance

Sodium chloride (NaCl) is produced when hydrochloric acid (HCl) reacts with sodium hydroxide (NaOH). Because both reactants are strong, the resulting salt is neutral ($\text{pH} \approx 7$). NaCl is the “common salt” used in cooking and food preservation. Large deposits of NaCl , called rock salt, are mined from ancient dried-up seas. Historically, NaCl symbolised Indian independence during Mahatma Gandhi’s Dandi March.

8. Activities

Identify Parent Acids and Bases for Given Salts (12 min)

Using the list of salts from Activity 2.14, students write the chemical formula of the acid and base that combine to form each salt, then group the salts into families based on common cations or anions.

Predict Salt Solution pH from Parent Strengths (14 min)

Students predict whether each salt from Activity 2.14 will be acidic, basic, or neutral by analysing the strength (strong/weak) of its parent acid and base, then compare predictions with the measured pH values.

9. Questions

- Q1. What is a salt family and give an example of two salts belonging to the same family.
- Q2. Predict whether a salt formed from a strong acid and a weak base will be acidic, neutral or basic, and explain why.
- Q3. In Activity 2.14, sodium acetate gave a pH of about 9. What does this indicate about the nature of the salt and its parent acid/base?
- Q4. State the pH value that defines a neutral solution and the ranges for acidic and basic solutions.
- Q5. Why is sodium chloride considered a neutral salt?